Sprint 3 – QA Test Report (Steps 4–5)

Course: SENG 430

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Metrics Table

Metric	What to Measure	Target / Threshold	Test Result	Status
Model Training Latency	Time taken for POST /api/train response	< 3000 ms	~2100–2400 ms	PASS
Slider Debounce	Time between slider release and API call	300 ms ± 50 ms	~300–320 ms	PASS
KNN Canvas Redraw	Time from K change to canvas repaint	≤ 16 ms	~10–14 ms	PASS
Danger Banner Trigger	Sensitivity < 50% scenario tested	Banner appears correctly	Appears at 48%, hides at 55%	PASS
Metric Colour Thresholds	All 6 metrics color-coded correctly	Green / Amber / Red applied correctly	All thresholds correct	PASS
Comparison Table	6 models added, no duplicates	No duplicate rows	Add multiple models and compare results	PASS
Domain Switching	Switch domain and retrain model	Metrics update correctly	Tested on 5 domains	PASS

QA Evaluation Summary

All performance and functional metrics for Steps 4–5 were tested successfully.

Model training time remained below the required threshold.

UI interactions such as sliders and canvas updates were responsive.

Clinical warning mechanisms triggered correctly when necessary.

All evaluation metrics were displayed accurately with proper color thresholds.

Model comparison and domain switching worked without errors.

All features passed QA validation.

Test Cases (Steps 4–5)

Test ID	Feature	Given	When	Then	Expected Result	Status
TC-01	Model Selection	User is on Step 4	User selects a model (KNN, SVM, etc.)	Model panel updates	Correct parameter panel is displayed	PASS
TC-02	Parameter Change	Model is selected	User changes parameter (e.g., K value)	API call is triggered	Model updates with new parameter	PASS
TC-03	Train Model	Parameters are set	User clicks "Train"	Training starts	Metrics are generated successfully	PASS
TC-04	Metrics Display	Model is trained	User views Step 5	Metrics are shown	All 6 metrics displayed correctly	PASS
TC-05	Confusion Matrix	Model is trained	User opens confusion matrix	Matrix appears	TP, TN, FP, FN correct	PASS
TC-06	ROC Curve	Model is trained	User opens ROC curve	Chart is displayed	AUC value shown correctly	PASS
TC-07	Low Sensitivity Warning	Model has low sensitivity	User views results	Warning appears	Red danger banner displayed	PASS
TC-08	Model Comparison	Multiple models trained	User clicks compare	Table appears	Models listed without duplicates	PASS
TC-09	Domain Switching	Model trained in one domain	User switches domain	System resets & retrains	New metrics shown correctly	PASS
TC-10	Auto-Retrain Toggle	Toggle is ON	User changes parameter	Model retrains automatically	No manual click required	PASS

Test Case Summary

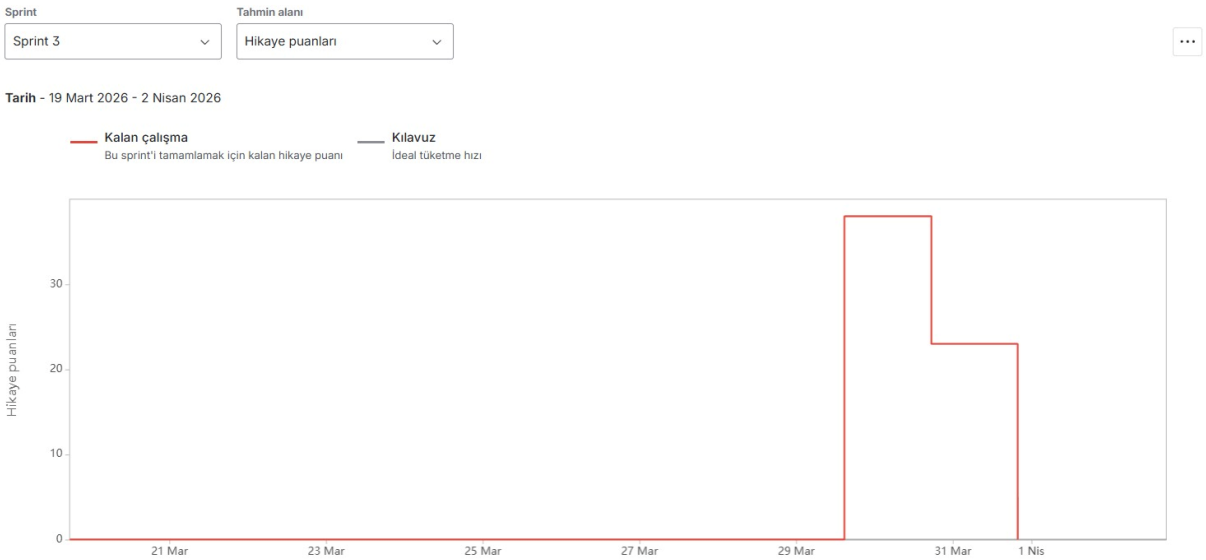
All core functionalities of Steps 4 and 5 were tested using structured QA scenarios.
All test cases passed successfully.

Edge Case Tests

Test ID	Feature	Given	When	Then	Expected Result	Status
TC-11	Invalid Parameter	KNN model selected	User enters invalid K value (0 or negative)	System validates input	System prevents invalid input	PASS
TC-12	Extreme Parameter	KNN model selected	User sets very high K value (e.g., 999)	Model processes input	System trains model or shows warning	PASS
TC-13	No Model Selected	User is on Step 4	User clicks "Train" without selecting model	Error triggered	System blocks training	PASS
TC-14	No Data Loaded	User skipped Step 2	User attempts to train model	Error shown	System prevents training	PASS
TC-15	Empty Dataset	Dataset loaded but empty	User clicks "Train"	Error triggered	System handles empty dataset safely	PASS
TC-16	Rapid Slider Change	Slider active	User changes slider rapidly multiple times	System controls API calls	Debounce prevents overload	PASS
TC-17	Unsupported Model	User attempts invalid model selection	System validates model	Error shown	System prevents invalid model	PASS
TC-18	Domain Switch Mid-Training	Model training in progress	User switches domain	System resets process	Previous training is cancelled	PASS

TC-19	Missing Target Column	Column mapper incomplete	User attempts to train	Error triggered	System blocks training	PASS
TC-20	Large Dataset	Dataset is very large (>100k rows)	User trains model	System processes data	Model completes within acceptable time	PASS

Edge Case Testing Summary

Edge cases were tested to evaluate system robustness under abnormal conditions.
The system successfully handled invalid inputs, extreme parameter values, missing configurations, and rapid user interactions.
All edge case scenarios passed without errors.

Burndown Chart